

IDTA Coursework 2



School of Computing

M33147-INTELLIGENT DATA AND TEXT ANALYTICS

Coursework: item 2

Date of Submission: 19/01/2026

Module Lecturers: Dr Atefeh Khazaei, Dr Ella Haig, Dr Alaa Mohasseb,
Dr Grace Golcarenenrenji

Student ID(s): up2102739 & up2110125 (Group AS and BG after
confirming we could work together from the start with Dr Atefeh Khazaei)

Word count: 2746

1.0 Data Processing	3
1.1removing punctuation	3
1.2removing numbers	3
1.3all text converted to lowercase	3
1.4Removing stop words	4
1.5Lemmatising	4
1.5.1Lemmatising with POS tags	4
1.6Stemming	5
1.7Preprocessing overview	5
2.0 Classification	6
2.1TF-IDF representation	6
2.2 Bag of Terms positive class results comparison	7
2.3 Bag of terms negative class results comparison	7
2.4 Bag of terms Accuracy and AUC comparison	8
2.5 ROC Curve analysis	9
3.0 BERT-Based model with fine-tuning	10
3.1 Model parameters	10
3.2 Model comparisons	10
3.3 Evaluation	11
4.0 Topic Detection	12
4.1 Algorithm used	12
4.2 results	12
4.2.1 Topics with large separation	13
4.2.2 Crossover	14
4.3 evaluation	14
5. References	15
6.Appendix A	15

1.0 Data Processing

This IMDB dataset contains 748 reviews. Before applying any classification algorithms, the text needed to be cleaned and standardised so the models could learn from meaningful patterns rather than superficial noise. The textual data was first processed to remove potential noise, with punctuation and numbers removed at the start. All text was then converted to lowercase to keep the wording consistent. Stop words were removed next to eliminate high-frequency terms that add little value to the classification task. Lemmatising and stemming followed to reduce words to their base or root forms. Together, these preprocessing steps ensure the dataset is cleaner, more compact, and more suitable for effective classification.

1.1 removing punctuation

Punctuation rarely carries meaningful information for most text classification tasks, particularly those focused on topic or sentiment. For example, “*Great!*” and “*Great*” should be interpreted identically by the model, as the exclamation mark does not alter the underlying semantic content. Retaining punctuation can also lead to unnecessary token variation, such as treating “*movie*” and “*movie,*” as separate features, which artificially expands the vocabulary and introduces noise into the representation. Removing punctuation therefore helps standardise tokens, reduce sparsity, and prevent the model from learning spurious distinctions based solely on punctuation marks. This simplification contributes to a cleaner feature space and supports more efficient and reliable downstream classification.

1.2 removing numbers

Numbers usually don’t contribute to meaning unless the task is numeric so an assumption was made that dates and prices are not meaningful in imdb reviews. In reviews, “10/10” or “3 stars” might matter, but assuming there is a different section for rating, this is unlikely to occur in the text. Therefore most digits are noise. This helps reduce vocabulary size and sparsity.

1.3 all text converted to lowercase

Converting all text to lowercase ensures consistency across the dataset by standardising the form of every token. Without this step, the model would treat “*Good*”, “*GOOD*”, and “*good*” as separate words, despite them carrying identical semantic meaning. This unnecessary variation inflates the vocabulary and introduces avoidable sparsity into the feature space. Lowercasing therefore helps consolidate equivalent terms, dramatically reducing vocabulary size and simplifying the representation. This not only cuts down processing but also prevents the classifier from learning artificial distinctions based solely on capitalisation, resulting in a cleaner and more efficient input for model training.

table 1.1. The first 5 reviews after punctuation & numbers were removed, and all text was made lowercase.

Steps taken	Text output [:5]
Removing punctuation/ removing numbers/ change text to lowercase	0,'a very very very slowmoving aimless movie about a distressed drifting young man ', 1,'not sure who was more lost the flat characters or the audience nearly half of whom walked out ', 2,'attempting artiness with black white and clever camera angles the movie disappointed became even more ridiculous as the acting was poor and the plot and lines almost nonexistent ', 3.'very little music or anything to speak of ', 4.'the best scene in the movie was when gerardo is trying to find a song that keeps running through his head '

1.4Removing stop words

Natural language is full of extremely common words that carry almost no discriminative power for classification. Words like *the*, *is*, *at*, *on*, and *but* appear in nearly every sentence regardless of sentiment, topic, or intent. Leaving them in inflates the vocabulary size, drowns out more meaningful terms, and introduces noise into sparse vector representations such as Bag-of-Words or TF-IDF. Removing these high-frequency, low-information words reduces dimensionality, sharpens the signal available to the model, and helps it focus on sentiment-bearing features. This not only speeds up training by reducing the number of features the classifier must process but also supports better generalisation by preventing the model from overfitting to common linguistic fillers.

1.5Lemmatising

Lemmatization reduces a word to its dictionary form (lemma) creating a more compact vocabulary, better grouping of related concepts, more meaningful features and less sparsity in vector space. When meaning matters within the textual data this is important as it preserves the true linguistic relationship between words.

1.5.1Lemmatising with POS tags

When POS tags are applied during lemmatisation, the lemmatiser takes into account the grammatical role of a word in a sentence. This is crucial because a word might have different base forms depending on its POS (e.g.'saw' as a verb (past tense of 'see') would lemmatize to 'see'.'saw' as a noun (the tool) would lemmatize to 'saw'.)

Incorporating these POS tags allows lemmatisation to be more accurate by finding the correct base form. Ultimately leading to a more semantically consistent representation of the text; improving the performance of text analysis.

Table1.2. A table showing the differences between lemmatising with/without POS tags

	Text output [:5]
Lemmatising	0 slowmoving aimless movie distressed drifting y... 1 sure lost flat character audience nearly half ... 2 attempting artiness black white clever camera ... 3 little music anything speak 4 best scene movie gerardo trying find song keep...
Lemmatising with POS tags	0 slowmoving aimless movie distressed drift youn... 1 sure lose flat character audience nearly half ... 2 attempt artiness black white clever camera ang... 3 little music anything speak 4 best scene movie gerardo try find song keep ru...

1.6 Stemming

Even after lemmatization, many words still have morphological variations that don't change meaning in a way that matters for classification. Stemming chops off the endings of words to reduce vocabulary, further collapsing related terms into a single form. This compression helps keep the feature space smaller and more manageable, which is especially useful for simple models that benefit from fewer dimensions and faster processing. By stripping words down to their core forms, stemming supports more efficient training while still preserving the essential meaning needed for accurate classification.

1.7 Preprocessing overview

Table1.3 A table showing the comparison of before and after text preprocessing.

	Text outputs[:5]
Before preprocessing	0,"A very, very, very slow-moving, aimless movie about a distressed, drifting young man. " 1,"Not sure who was more lost - the flat characters or the audience, nearly half of whom walked out. " 2,"Attempting artiness with black & white and clever camera angles, the movie disappointed - became even more ridiculous - as the acting was poor and the plot and lines almost non-existent. " 3,Very little music or anything to speak of. 4,The best scene in the movie was when Gerardo is trying to find a song that keeps running through his head.
After preprocessing	0, slowmoving aimless movie distressed drift young man 1, sure lose flat character audience nearly half walk 2, attempt artiness black white clever camera angle movie disappointed become even ridiculous act poor plot line almost nonexistent

	3, little music anything speak
	4, best scene movie gerardo try find song keep run head

Table 1.4 A table showing what noise was reduced from each preprocessing step.

Preprocess steps	Noise reduced
Removing punctuation	• Token fragmentation • Artificial vocabulary inflation
Removing numbers	• Irrelevant numeric tokens • Random IDs, years, product codes
Change text to lowercase	• Case-based duplicates • Artificial distinctions between identical words
Removing stop words	• Extremely common words that add no meaning • High-frequency filler terms (the, is, and, but) • Words that inflate vocabulary without improving classification • Tokens that blur differences between classes
Lemmatising	• Inflectional variation (run, runs, running all to run) • Redundant word forms that represent the same concept • Artificial distinctions caused by tense, number, or comparison • Extra vocabulary created by grammatical changes
Stemming	• Morphological endings (-ing, -ed, -ness, -tion) • Surface-level variations that don't change meaning for ML • Excess vocabulary created by word-form complexity • Irrelevant suffixes that increase sparsity

2.0 Classification

2.1 TF-IDF representation

TF-IDF was selected as the primary text representation method because it provides a more discriminative and information-rich space than a standard Bag-of-Words model. While Bag-of-Words relies only on raw term frequencies, it treats all words as equally informative, which leads to high-frequency but semantically weak terms dominating the representation. But, TF-IDF down weights ubiquitous words and highlights terms that are distinctive within individual documents. This weighting mechanism is advantageous for sentiment

classification on IMDB reviews, where emotionally connotative adjectives (e.g amazing, terrible, boring, brilliant) carry far more predictive value than structural or filler words.

By reducing the influence of common but uninformative tokens and highlighting sentiment-bearing vocabulary, TF-IDF produces a matrix with improved signal-to-noise characteristics[1]. This directly benefits linear classifiers such as Logistic Regression, SVMs, and Naive Bayes, which rely on meaningful variation in feature weights to separate positive and negative sentiment. Additionally, TF-IDF produces a more compact and efficient representation and reduces the risk of overfitting by preventing frequent neutral terms from overwhelming the model[1]. Overall, TF-IDF yield is more balanced, interpretable, and discriminative representation, leading to stronger classification performance on the IMDB dataset compared with Bag-of-Words counts.

2.2 Bag of Terms positive class results comparison

Table 2.1 Positive class metric results using TF-IDF representation.

Model	Precision	Recall	F1-score
Decision Tree	0.72	0.74	0.74
Naive Bayes	0.84	0.79	0.82
SVM	0.86	0.80	0.83

The Decision Tree model performs the weakest overall, with precision, recall, and F1-score all in the mid-0.70 range. This reflects the difficulty tree-based methods face when operating in high-dimensional, sparse spaces such as TF-IDF; overfitting local patterns rather than capturing broader term-weight relationships. In contrast, Naive Bayes shows a marked improvement, achieving higher precision(0.84) and a stronger F1-score(0.82). Its probabilistic structure aligns well with TF-IDF representations, allowing it to exploit term-frequency patterns effectively, although its recall is slightly lower due to the independence assumption limiting its ability to handle subtle emotional distinctions in the text.

The strongest performance is achieved by the SVM classifier, which records the highest precision (0.86), recall (0.80), and F1-score (0.83). SVMs are well suited to text classification because they handle sparse, high-dimensional feature spaces robustly and maximise class separation through margin optimisation. Overall, SVM performs best, followed by Naive Bayes, with Decision Trees behind due to structural limitations in TF-IDF representations.

2.3 Bag of terms negative class results comparison

Table 2.2 negative class metric results using TF-IDF representation.

Model	Precision	Recall	F1-score
Decision Tree	0.69	0.67	0.68
Naive Bayes	0.77	0.83	0.80

SVM	0.78	0.84	0.81
-----	------	------	------

The Decision Tree model performs the weakest, with precision (0.69), recall (0.67), and F1-score (0.68) all notably lower than the other classifiers. This reflects the structural limitations of the tree-based method; its tendency to overfit local partitions of the feature space, plus its struggle to generalise sentiment patterns that are distributed across many weighted terms.

Naive Bayes(NB) demonstrates a substantial improvement, achieving higher precision (0.77) and the strongest recall (0.83) among the three models. This indicates that NB is particularly effective at identifying negative reviews, possibly due to its probabilistic treatment of term distributions, working well with TF-IDF's weighting of informative words. The resulting F1-score of 0.80 shows that NB maintains a strong balance between avoiding false positives and capturing a broad portion of the negative class.

SVM delivers the best performance, with precision (0.78), recall (0.84), and F1-score (0.81) all marginally higher than Naive Bayes. Its margin-based optimization enables it to separate classes effectively in sparse, high-dimensional TF-IDF spaces, leading to robust generalisation. Overall, SVM and Naive Bayes both perform strongly, but SVM maintains a slight advantage, while the Decision Tree lags significantly behind.

2.4 Bag of terms Accuracy and AUC comparison

Table 2.3 accuracy and AUC results using TF-IDF representation.

Model	Accuracy	AUC
Decision Tree	0.703	0.693
Naive Bayes	0.808	0.781
SVM	0.821	0.820

The accuracy and AUC results in table2.3 show a clear performance gradient across the three classifiers when using TF-IDF features. The Decision Tree achieves the lowest scores (accuracy 0.703, AUC 0.693), reflecting its difficulty in modelling with TF-IDF. Naive Bayes performs substantially better, with an accuracy of 0.808 and an AUC of 0.781, indicating stronger overall differentiation between positive and negative reviews. Its probabilistic framework aligns well with TF-IDF weighting, enabling it to separate classes more reliably than the tree-based model.

SVM delivers the strongest results, achieving the highest accuracy (0.821) and the highest AUC (0.820). These metrics demonstrate that SVM not only classifies individual reviews the most successfully overall but also preserves a higher quality of class separation at all decision thresholds.

2.5 ROC Curve analysis

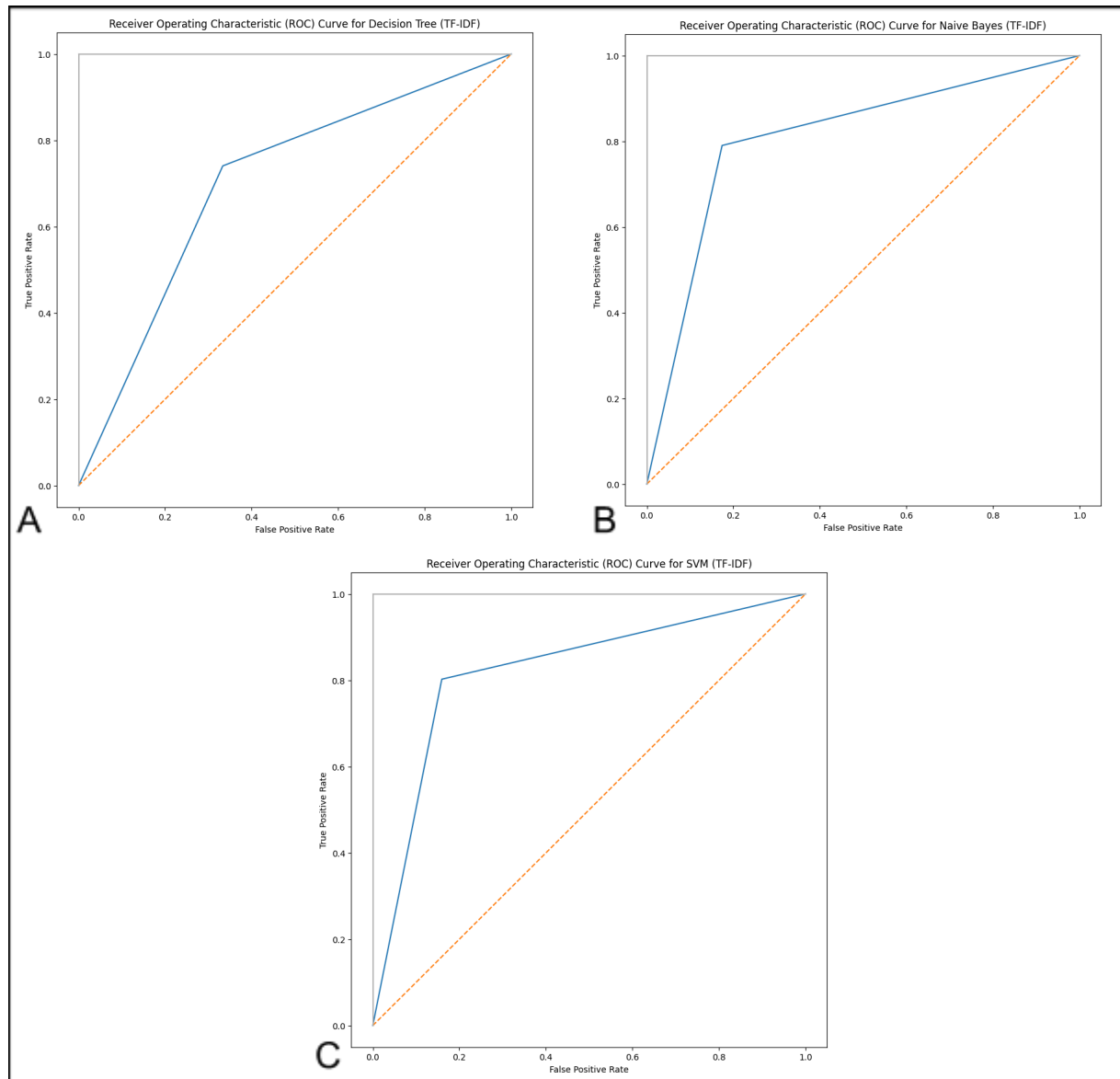


Fig 2.1 Comparison of the Receiver Operating Characteristic (ROC) curve for all classification algorithms using TF-IDF representation. A) for Decision Tree. B) for Naive Bayes. C) for SVM.

The ROC curves in Fig 2.1 reinforce the performance differences observed in the quantitative metrics. The Decision Tree shows the weakest curve, staying closer to the $x=y$ baseline and indicating limited ability to distinguish between positive and negative reviews. In contrast, the Naive Bayes curve initially rises more sharply toward the top-left corner, reflecting stronger class-separation, aligning with its higher recall and F1-scores. The SVM curve slightly demonstrates the most favourable shape, consistently dominating the other models across the full range of false-positive rates. Its biggest gradient on the lefthand side of the graph corresponds directly with its highest AUC value. Overall, the ROC curves visually confirm the earlier results: SVM provides the highest quality of classification performance, followed by Naive Bayes, with the Decision Tree trailing behind.

3.0 BERT-Based model with fine-tuning

3.1 Model parameters

For this task a pre-trained Bert model (Bert en encased) was chosen due to its ability to provide quality results while remaining relatively fast and being fine tunable. The text was first processed using the corresponding bert pre-processor before being passed to a pre-trained transformer encoder with all parameters being set to trainable. The model was optimised using AdamW with a low learning rate and warm-up schedule to ensure stable fine tuning by preventing large updates early on in training. Binary cross-entropy was used with accuracy and AUC in order to assess the quality of the model.

3.2 Model comparisons

Table 3.1 results from all classification (0) indicating negative class and (1) indicating positive class

Model	Accuracy	Precision	Recall	F1-score	ROC/AUC	Loss
Bert	0.893	(0) 0.84 (1) 0.96	(0) 0.96 (1) 0.83	(0) 0.90 (1) 0.89	0.857	0.323
Decision Tree	0.706	(0) 0.69 (1) 0.72	(0) 0.67 (1) 0.74	(0) 0.68 (1) 0.73	0.703	N/A
Naive Bayes	0.806	(0) 0.77 (1) 0.84	(0) 0.83 (1) 0.79	(0) 0.80 (1) 0.82	0.808	N/A
SVM	0.820	(0) 0.78 (1) 0.86	(0) 0.84 (1) 0.80	(0) 0.81 (1) 0.83	0.821	N/A

Across the different models the fine-tuned BERT-based classifier performed best across all metrics. It is observed in Table 4.1 to have achieved the strongest performance across specifically ROC/AUC and accuracy compared to the classic models. This indicates a contextual approach is more efficient for sentiment based classification where text is informal and has more nuance.

The models trained using “bag of terms”, especially decision trees, struggle to find meaningful connections between language likely due to the informal and irregular language used throughout reviews. Bert approached this by learning context to words provided in the training set over multiple generations and applies this context to unseen data.

The confusion matrix seen in Fig 3.1 illustrates the performance of the BERT-based classification model on the test set. The model correctly classified most negative and positive items with only a few misclassifications present. The slight disparity between false

negative and false positive predictions suggests a reduced sensitivity to positive sentiments in review text. The confusion matrix reinforces BERT-based modelling as a strong classifying choice for this task. Additionally a loss value of 0.323 indicates the model is making confident predictions when it is correct and remaining cautious when incorrect.

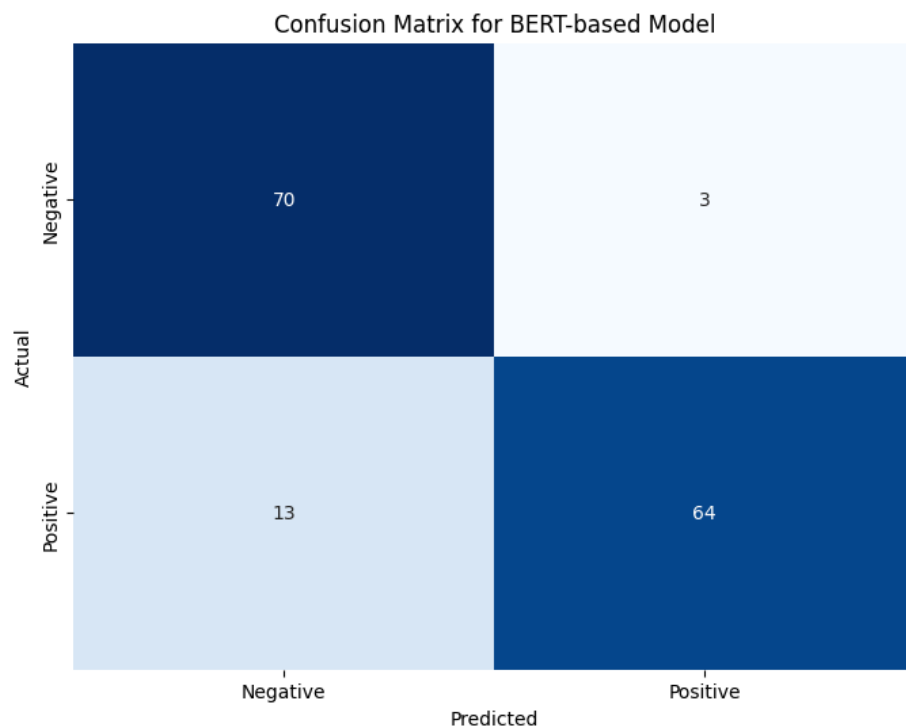


Figure 3.1 Confusion matrix from BERT-based classification

3.3 Evaluation

Despite strong performance BERT-Based modelling does have drawbacks, mainly due to the introduction of additional computational complexity. Training fine-tuned BERT-based models required a GPU which reduces the availability of results and vastly increases training times. This drawback may limit the usage of BERT when in resource constrained environments as the improvements seen in Table 3.1 are not significant enough to warrant purchase of extra components due them being incremental improvements rather than transformative.

Despite these limitations when the computational power and time is available, fine tuned BERT-based modelling constantly achieves superior results when being compared to classic classification models. This alone makes them suitable for when accuracy is prioritised over computational or time efficiency.

4.0 Topic Detection

4.1 Algorithm used

Latent Dirichlet Allocation(LDA) was used in order to generate topic detection for the IMDB dataset. LDA was used due to its good trade-off delivering a high level of explainability and

not sacrificing strong performance as a result. The interaction with tools such as LDAvis allows for qualitative analysis of all topic coherence and separability.

The number of topics was set for the task at 10, due to this a model with good interpretability was required to be able to assess potential overlap in topics. Other models such as bert based neural topics models may have been able to achieve more separated results but this comes at the cost of readability. A coherent based selection may have yielded fewer results with less overlap and should be tested to further improve results. LDA does come with limitations associated with bag of words assumptions which could result in limited separation between closely related topics.

4.2 results

Fig 4.1 demonstrates the intertopic map showing the distance, separation and size of each of the 10 topics generated by LDA. There is a significant overlap in 7 of the topics in the bottom right quartile of the map. However there are 3 distinct topics with a large separation between the main cluster of crossover which suggest niche topics. The context of each topic can be inferred from relevant terms on the right hand side of Fig 4.1.

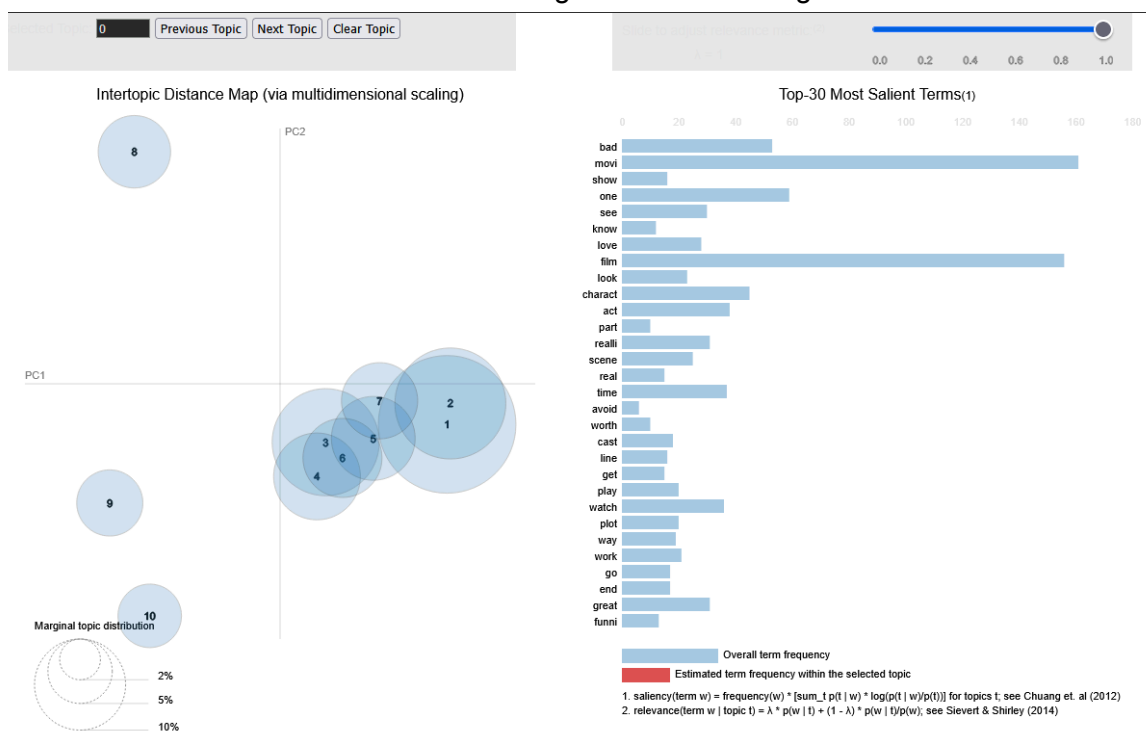


Figure 4.1 Results of topic detection

Table 4.1 Results of topic detection per topic

Topic	% Tokens	Top Keywords	Interpreted Theme	Assessment
1	22.8%	film, movie, like, good, bad, character, love, disappoint	General Movie Opinion / Personal Reaction	Very broad and dominant topic capturing subjective viewer opinions and mixed sentiment.
2	14.8%	film, bad, movie, scene, plot, story, watch, good, worst	Plot and Scene Evaluation	Coherent evaluative topic with a clearer focus on narrative

				elements such as plot and scenes.
3	13.8%	movie, film, character, cast, watch, write, bad	Character and Performance Evaluation	Focuses on characters, cast, and writing quality alongside evaluative language.
4	9.1%	movie, film, watch, recommend, great, actor	Recommendation and Viewing Experience	Emphasises enjoyment and whether films are worth recommending.
6	7.5%	movie, film, scene, script, plot, good, bad	Script and Story Engagement	Captures discussion of screenplay quality and narrative engagement.
7	7.0%	movie, film, character, actor, good, great, performance	Acting and Performance Evaluation	Focuses on actor performances and character portrayal, with positive and negative evaluative language.
8	6.3%	good, great, best, excellent, style	Positive overall evaluation	Coherent and well-separated; clearly captures positive sentiment
9	5.3%	bad, worth, watch, time, classic	Mixed/negative reflection	Distinct evaluative topic; slightly noisy due to mixed sentiment
10	4.9%	character, actor, feel, love, worth	Character-focused audience reaction	Clear emotional focus with mild overlap with general review language

4.2.1 Topics with large separation

Topic 8 is separated from the main cluster indicating a clear niche topic of coverage for the review. By observing the relevant terms present in Fig 4.2 a conclusion can be drawn that figure 8 describes overall positive reviews of movies. Evaluative language such as *good*, *great* and *best* indicates positive reviews with references to *plot*, *total* and *style* demonstrate that the reviews are likely addressing an overall coverage rather than specific choices. A presence of 6.3% of tokens suggests the topic is consistently present but not dominant in the dataset.

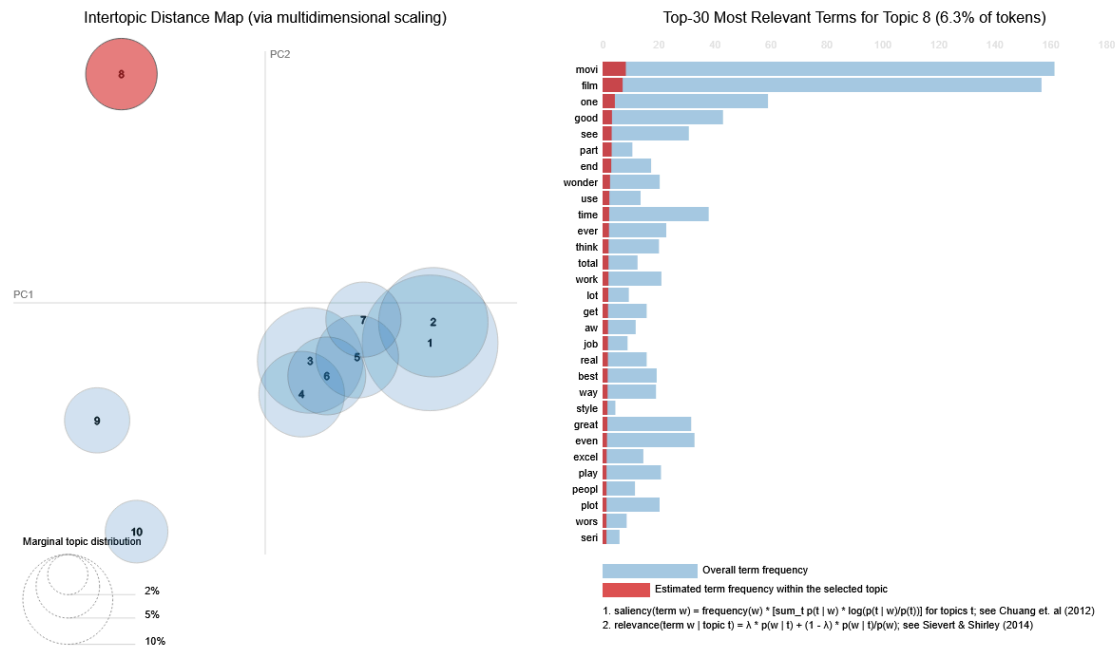


Figure 4.2 review of topic 8 relevant terms

Topic 9 and 10 are also largely separated from the main cluster which indicates niche and specific topics. As observed in Table 4.1 topic 9 reviews more negative or mixed opinions and tends to be more relevant to old movies, reviewing with retrospective, this is observed through keywords *Classic* and *old*. This differs from topic 8 where the reviews were all positive suggesting that negative and positive reviews can be largely separated based on emotive keywords.

Topic 9 interestingly is separate to this binary separation of “good” or “bad” where topic 9 focuses largely on reviews of characters and actors. Keywords seen in Table 4.1 describe emotive responses towards the characters instead of the movie or show overall. The smaller token size suggests while these reviews are present and easy to discern they don’t appear dominant or common amongst the dataset.

4.2.2 Crossover

There is a large crossover between topics 1-7 although each topic still has an individual topic of coverage. As can be observed from Table 4.1 a large impact of the crossover was the relevance of keywords such as *movies* and *film* which are less common in topics 8-10. Additionally topics 8-10 review emotive feelings from the viewer about the overall experience and characters whereas topics 1-7 are focusing on specific attributes of the movie such as script, acting or plot. These keywords tend to co-concur in one review leading to the crossover between topics.

4.3 evaluation

Overall topic detection demonstrated coherent results and interpretability especially in topics with emotional or evaluative responses (topics 8-10). However there was significant overlap in topics 1-7 due to common keywords and co-currence of topics through reviews. The

presence of the dominant background topic (topic 1) also contributes to a reduced separation between topics. This indicates that 10 topics could have been too many to generate consistent meaningful topics.

Despite the limitation of crossover topics the LDA still provides useful insight into the dataset distinguishing content-based reviews and overall viewer sentiment. The use of 10 topics may have negatively impacting separation by forcing the model to find unique topics where they did not exist. For example topic 2 is almost completely crossed over with topic 1 suggesting they would be better as a combined topic.

5. References

[1] Das, M., Kamalanathan, S., & Pja, A. (2023). A comparative study on TF-IDF feature weighting method and its analysis using unstructured dataset.
<https://doi.org/10.48550/arXiv.2308.04037>

6. Appendix A

Contribution table:

ID	Contribution	Tasks
Up2102739	50%	Section 1, Section 2/3.
Up2110125	50%	Section 2/3, section 4